Benchmark: **MA.7.DP.2.3** · Find theoretical probability of an event Student code: _____ Calculator: yes

Learning goal: I can find the theoretical probability of an event. (circle your level after scoring)

4	I can find the probability of an event described by a condition (such as 'an even number' or 'not red') and explain my count.
3	I can find the theoretical probability of a simple event and write it as a fraction, decimal, or percent.
2	I can count the favorable outcomes and the total outcomes.
1	I need help to start this benchmark.

1. A fair 20-sided die has faces numbered 1 through 20. The die is rolled once.

What is the probability of rolling a number greater than 15? Write your answer as a fraction in simplest form.

2. A fair spinner has 25 equal sections. 6 sections are green. The spinner is spun once.

What is the probability that the spinner lands on green? Write your answer as a decimal.

Turn over →

Benchmark: **MA.7.DP.2.3** · Find theoretical probability of an event Student code: _____

3. A bag holds 12 marbles, and 3 of them are red. One marble is drawn at random.

What is the probability of drawing a red marble?

A. $\frac{3}{11}$

B. $\frac{1}{4}$

C. $\frac{3}{4}$

D. $\frac{1}{6}$

4. Letter cards for the word MEASUREMENT are placed in a bag, one card for each letter. One card is drawn at random.

What is the probability that the card shows the letter M? Write your answer as a fraction in simplest form.

5. A standard deck has 52 cards: 13 cards (ace, 2 through 10, jack, queen and king) in each of four suits — hearts and diamonds, which are red, and clubs and spades, which are black. One card is drawn at random.

What is the probability of drawing a card that is not a black king? Write your answer as a fraction in simplest form.

Score: ____ / 5 **My level: 1 2 3 4** **→ copy both to your progress strip**

Answer key — C-DP.2.3 · MA.7.DP.2.3 · Find theoretical probability of an event

Unit 2: Probability. Regular = on-benchmark item; Rigor = the item that needs the benchmark's reasoning. Mastery on the weekly check = 4 of 5.

Weekly check (5 items)

1. Regular $\frac{1}{4}$

The numbers greater than 15 are 16 through 20: 5 favorable outcomes out of 20.

$P = \frac{5}{20} = \frac{1}{4}$.

2. Regular **0.24**

All 25 sections are equally likely, so the sample space has 25 outcomes and 6 of them are green.

$P(\text{green}) = \frac{6}{25}$.

$6 \div 25 = 0.24$.

3. Regular **B.** $\frac{1}{4}$

$P(\text{red}) = \text{ favorable outcomes} \div \text{ total outcomes} = \frac{3}{12} = \frac{1}{4}$.

4. Regular $\frac{2}{11}$

There are 11 cards, one for each letter of MEASUREMENT, and each card is equally likely to be drawn.

The letter M is on 2 of the cards: 2 favorable outcomes out of 11.

$P = \frac{2}{11}$.

5. Rigor $\frac{25}{26}$

Only 2 cards are black kings: the kings that are also black. Every other card is not a black king: $52 - 2 = 50$ favorable cards.

$P = \frac{50}{52} = \frac{25}{26}$.

Re-check strip (3 items, C-DP.2.3-R)

1. Regular **75%**

All 4 sections are equally likely, so the sample space has 4 outcomes and 3 of them are yellow.

$P(\text{yellow}) = \frac{3}{4}$.

$3 \div 4 = 0.75$, and $0.75 = 75\%$.

2. Regular $\frac{3}{8}$

The numbers greater than 5 are 6 through 8: 3 favorable outcomes out of 8.

$P = \frac{3}{8}$.

3. Regular **B.** $\frac{2}{9}$

$P(\text{red}) = \text{ favorable outcomes} \div \text{ total outcomes} = \frac{4}{18} = \frac{2}{9}$.

Comparison strip (2 items, C-DP.2.3-X) — used on a later week's re-check when this benchmark was *not* retaught

1. Regular **B.** $\frac{1}{6}$

$P(\text{red}) = \text{ favorable outcomes} \div \text{ total outcomes} = \frac{2}{12} = \frac{1}{6}$.

2. Regular **0.75**

All 8 sections are equally likely, so the sample space has 8 outcomes and 6 of them are orange.

$P(\text{orange}) = \frac{6}{8} = \frac{3}{4}$.

$6 \div 8 = 0.75$.

Reteach plan — C-DP.2.3

MA.7.DP.2.3 · Find theoretical probability of an event

Item that gave the most trouble (from both tally sheets):

What the mistake looks like:

Worked example for Tuesday (8–10 min):

Three practice items (one change at a time):

Decision line: Reteach on _____ Huddle: in person / paper Date: _____ Re-check given on _____

7th Grade Math · Re-check (3 items) C-DP.2.3-R

Benchmark: **MA.7.DP.2.3** · Find theoretical probability of an event Student code: _____ Calculator: yes

1. A fair spinner has 4 equal sections. 3 sections are yellow. The spinner is spun once.

What is the probability that the spinner lands on yellow? Write your answer as a percent.

2. A fair 8-sided die has faces numbered 1 through 8. The die is rolled once.

What is the probability of rolling a number greater than 5? Write your answer as a fraction in simplest form.

3. A bag holds 18 marbles, and 4 of them are red. One marble is drawn at random.

What is the probability of drawing a red marble?

A. $\frac{4}{17}$

B. $\frac{2}{9}$

C. $\frac{7}{9}$

D. $\frac{1}{6}$

Score: ____ / 3 → **copy to your progress strip**

7th Grade Math · Re-check (3 items) C-DP.2.3-R

Benchmark: **MA.7.DP.2.3** · Find theoretical probability of an event Student code: _____ Calculator: yes

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Score: ____ / 3 → **copy to your progress strip**

7th Grade Math · Comparison items (2) C-DP.2.3-X

Benchmark: **MA.7.DP.2.3** · Find theoretical probability of an event Student code: _____ Calculator: yes

1. A bag holds 12 marbles, and 2 of them are red. One marble is drawn at random.

What is the probability of drawing a red marble?

A. $\frac{5}{6}$

B. $\frac{1}{6}$

C. $\frac{2}{11}$

D. $\frac{1}{12}$

2. A fair spinner has 8 equal sections. 6 sections are orange. The spinner is spun once.

What is the probability that the spinner lands on orange? Write your answer as a decimal.

Score: ____ **/ 2** → **copy to your progress strip**

7th Grade Math · Comparison items (2) C-DP.2.3-X

Benchmark: **MA.7.DP.2.3** · Find theoretical probability of an event Student code: _____ Calculator: yes

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C. $\frac{2}{11}$

D. $\frac{1}{12}$

2. A fair spinner has 8 equal sections. 6 sections are orange. The spinner is spun once.

What is the probability that the spinner lands on orange? Write your answer as a decimal.

Score: ____ **/ 2** → **copy to your progress strip**